High Performance Computing
OPTIMIZATION Outline

Luca Tornatore - I.N.A.F.

DATA SCIENCE & ARTIFICIAL INTELLIGENCE

SCIENTIFIC & DATA-INTENSIVE COMPUTING

2025-2026 @ Università di Trieste

Outline

Basic traits of modern CPU architecture

First steps in optimization
Caveat on floating-point arithmetic

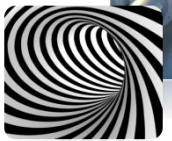
Running model,
Heap & Stack

Some
preliminary stuff on
Compiler

The cache hierarchy and
how to write
cache-friendly code

Introduction &
general
concepts
on optimization

The impact of
conditional evaluation on
code performance



Pipelined instructions and
How to optimize loops

that’s all, have fun